

3.2.1 Types of Functions (Built-in vs User-defined)

بلٹ ان فنکشن پائٹھن میں پہلے سے موجود ہوتے ہیں، جیسے `len()` اور `print()`۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 10 of 17

3.2.1 Types of Functions (Built-in vs User-defined)

LEARN IT

There are two main types of functions in Python.

a) Built-in functions are already defined by Python. You just call them. Examples you have already used: `len()`, `range()`, `sum()`, `print()`, `input()`, `min()`, `max()`, `sorted()`, `int()`, `float()`, `type()`.

b) User-defined functions are functions **you** create with `def` for tasks that no built-in covers in one step — greeting a user, dividing with a zero check, writing a grade file.

	Built-in	User-defined
Who writes it?	The Python language	The programmer
How you start	Just call the name	First write <code>def name(...):</code>
Examples	<code>len("Ali") → 3</code>	<code>def add(a, b): return a + b</code>
Purpose	Everyday operations needed by everyone	Your program's specific jobs

WORKED EXAMPLE

Side by side. Counting letters is built-in. Combining a greeting with a name is user-defined.

```
print(len("Sindh"))

def greet_student(name):
    print("Hello,", name)

greet_student("Hassan")
greet_student("Malika")
```

OUTPUT

```
5
Hello, Hassan
Hello, Malika
```

You did not write the code inside `len`. You did write the code inside `greet_student`, then reused it for two names.

REMEMBER IT

If Python already has it (`len`, `sum`, `max`), do not rewrite it. Write a user-defined function when you are packaging **your** steps.

COMMON MISTAKE

Do not name your function `list`, `print`, `sum` or `input`. You would hide the built-in name for the rest of that file. Choose names such as `print_bill` or `sum_marks`.

اردو میں سمجھیں

بلٹ ان فنکشن پائتھن میں پہلے سے موجود ہوتے ہیں، جیسے `len()` اور `print()`۔ یوزر ڈیفائنڈ فنکشن وہ ہیں جو پروگرامر `def` سے خود بناتا ہے تاکہ مخصوص کام بار بار کیا جا سکے۔ جو کام بلٹ ان پہلے کر دیتے ہیں انہیں دوبارہ نہ لکھیں۔

CHECK YOURSELF

1. Is `range()` built-in or user-defined?
2. Which keyword creates a user-defined function?

Answers: 1. Built-in. 2. `def`.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Built-in functions include:

- a. len, print, min (provided by Python)
- b. Only ones you def
- c. Only pip packages
- d. Only SQL

MCQ 2. User-defined functions are:

- a. Ones you write with def
- b. print and len
- c. CPU instructions
- d. OSI layers

MCQ 3. You still:

- a. Call both with name()
- b. Cannot call built-ins
- c. Must recompile C
- d. Must draw Figma

MCQ 4. A module function like math.sqrt is:

- a. Library/built-in-standard, not your def
- b. A user def in your file
- c. A dict
- d. A stack

MCQ 5. Why write your own if print exists?

- a. Your problem's chunks (bill tax, grade band) are not built in
- b. Python forbids print
- c. Built-ins cannot be called
- d. def is illegal

MCQ 6. Help(len) documents a:

- a. Built-in
- b. Your secret function always
- c. A table
- d. A query

Extra short questions

1. Differentiate built-in vs user-defined with examples.
2. Can user-defined call built-in?
3. Where do library functions sit?
4. One exam trap.

Extra long question

1. In one program use a built-in, a standard-library function, and a user-defined function. Label each and show output.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Built-in: print, len. User: def grade(m): ...
2. Yes — that is normal.
3. Neither in your file nor mysterious — imported modules.
4. Calling a function you never defined and is not built-in → NameError.

Long

1. print/len; math.sqrt; def mean. Three labels in comments. Exact numeric output.

Next lecture: 3.2.2 Functions with Return Values